Bottom vector clock: $\bot = (0, 0, \dots, 0)$. Every thread has logical clock 0

Partial order on vector clocks: $V_1 \sqsubseteq V_2$ if and only if $\forall t\,.\, V_1(t) \leq V_2(t)$. One vector clock is "less-than or equal to" another vector clock if logical clocks are pointwise less-than or equal.

Join of vector clocks: $V_1 \sqcup V_2$ = pointwise maximum of V_1, V_2

Increment: $inc_t(V) = V[t \ \mapsto \ V(t) + 1]$. Same as V, but logical clock for t is incremented.